

BAX 423 Homework 1 Part 4

NOAA Storm Events: EF2+ Tornadoes 2020-2025

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Data source: NCEI Storm Events Database detail CSV files. Filter: EVENT_TYPE Tornado, TOR_F_SCALE in EF2 through EF5 plus rare legacy F2 through F5. Tie-break for peak month per state: smallest month number if tied. Years 2020 through 2025.

Summary counts

Metric	Value
All-event rows loaded all types all years	410,192
EF2+ tornado rows analysis set	1,164
Total deaths DEATHS_DIRECT plus DEATHS_INDIRECT	396
Fatal incident rows with deaths greater than zero	137
Percent of fatal narratives mentioning mobile home or trailer	50.36% rows 69

States with the most EF2+ tornadoes top 15

State	Count
MISSISSIPPI	112
TEXAS	111
ALABAMA	93
IOWA	74
TENNESSEE	61
ARKANSAS	61
MISSOURI	60
LOUISIANA	53
ILLINOIS	52
OKLAHOMA	50
KENTUCKY	45
NEBRASKA	40
GEORGIA	39

INDIANA	37
FLORIDA	36

Frequency by month all states combined

Month	Count
1	68
2	43
3	254
4	242
5	178
6	81
7	51
8	37
9	12
10	30
11	31
12	137

States where May is the peak month for EF2+ counts

States where for that state alone May had the highest count. Ties use earliest month. Use this list for the course email part 4e.

COLORADO, KANSAS, MICHIGAN, MINNESOTA, TEXAS, VIRGINIA

Regional pattern

Among the highest-frequency states, many are Great Plains, Dixie Alley, or Midwest states where spring and early-summer severe weather is climatologically common. Coastal and western states typically rank lower in raw tornado counts, consistent with the central and southeastern U.S. tornado maxima.

Examples Plains Midwest Southeast among top 10 states: MISSISSIPPI, TEXAS, ALABAMA, IOWA, TENNESSEE, ARKANSAS, MISSOURI, LOUISIANA, ILLINOIS, OKLAHOMA.

Part 4(e) email draft

To: cmdemena@ucdavis.edu

Subject: BAX423 Tornado Watch

Body: Tell me to avoid these states in the month of May: COLORADO, KANSAS, MICHIGAN, MINNESOTA, TEXAS, VIRGINIA.

Sign off: From Logan Garcia & Bonnie Hines

Send from a throwaway non-university inbox (we used Proton Mail; syllabus suggested Gmail).

Generated for BAX 423 HW1. NOAA data are public domain NCEI.